SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	"Hack into Google, No mistake"
Project / Solution Name	SOPBox: Smart Receiving Check for F&B SMEs
Name (Leader)	Lye Wei Lun
Email & Phone (Leader)	dellralllye@gmail.com & 018-960 6378
University Affiliation(s)	Tunku Abdul Rahman University of Management and Technology (TAR UMT)
Live Prototype URL	https://sopbox.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Dellrall/SOPBox

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SOPBox is a phone-first receiving-control platform built with React and TypeScript on the frontend and Python FastAPI on the backend. The application is deployed on Replit Autoscale, with PostgreSQL used to store business workspaces, user accounts, products, suppliers, purchase orders, deliveries, stock records and audit logs. PostgreSQL Row Level Security (RLS) helps isolate each business's data.

For delivery-order image extraction, the backend sends images to a RunPod Serverless endpoint running vLLM with the pre-trained Qwen3-VL-8B-Instruct-FP8 model. A second OpenAI-compatible AI service supports sales-report reading and the assistant. AI responses are validated by the backend before use. Deterministic application rules check temperatures, quantities, pack conversions and purchase-order balances. Staff confirm extracted information, while managers handle exceptions through the approval workflow.

Telegram Bot API supports notifications and bot interactions. Google OAuth and WebAuthn support optional sign-in methods.

The core principle will be AI reads. Rules decide. Humans approve.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12 (Replit runtime; supports 3.11+)	Backend programming language	PSF License
FastAPI >=0.110	Backend REST API	MIT
Uvicorn >=0.29	ASGI application server	BSD-3-Clause
Pydantic >=2.6	Data validation and request/response models	MIT
React ^18.3.1	Frontend user interface	MIT
React DOM ^18.3.1	React browser rendering	MIT
TypeScript ^5.5.0	Typed frontend development	Apache-2.0
Vite ^7.0.0	Frontend development and build tool	MIT
Tailwind CSS ^4.0.0	Frontend styling	MIT
PostgreSQL 16 (Replit)	Relational database and workspace data isolation	PostgreSQL License
psycopg[binary] >=3.2	PostgreSQL database connectivity	LGPL-3.0-or-later
openpyxl >=3.1	Spreadsheet processing	MIT
SimpleWebAuthn Browser ~14.0.0 and Python WebAuthn >=2.5,<3	Passkey registration and authentication	MIT
Playwright ^1.64.1	Automated frontend end-to-end testing	Apache-2.0
vLLM 0.30 (RunPod worker image v2.28.0)	Serves the vision-language model	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API	Sends delivery-order and other supported images to the project's hosted vision-language model for extraction	Paid usage-based / credits; sponsor-provided only if officially granted
KrackedDev OpenAI-compatible AI service (LiteLLM endpoint)	Reads sales reports and supports the AI assistant	API key / provider-dependent usage
Telegram Bot API	ends delivery alerts, supports Telegram sign-in and handles bot and team-group interactions	Free
Google OAuth / OpenID Connect	Optional Google sign-in and account linking	Browser API; not a paid external AI API
WebAuthn browser API	Passkey registration and authentication using supported device authenticators	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts and deploys the SOPBox frontend and FastAPI backend through one application process	Replit core
Replit PostgreSQL	Stores user accounts, workspaces, products, suppliers, purchase orders, deliveries, stock, sales and audit records	Replit PostgreSQL; PostgreSQL 16
RunPod Serverless	Runs the vLLM worker and Qwen3-VL-8B-Instruct-FP8 model for AI image extraction	Serverless endpoint; GPU pool documented as AMPERE_24; verify current configuration
Github	Source-code hosting, version control and automated testing workflows	Public Repository: SDCIdea26
Telegram	Delivers configured notifications and supports bot interactions	Telegram Bot API

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3-VL-8B-Instruct-FP8	Qwen model served through vLLM on RunPod Serverless	Pre-trained inference for extracting product names, quantities and units from delivery-order images; no fine-tuning declared	Apache-2.0
GPT-5.4-mini (configured default)	KrackedDev OpenAI-compatible service through https://litellm.rapidscreen.io/v1	Pre-trained hosted inference for sales-report reading and the AI assistant	Provider/service terms; not a self-trained model

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
SOPBox business and operational records	Created by users in the SOPBox application	Synthetic	Team-created; no third-party dataset licence claimed	N
SOPBox test and demonstration data	Created for project development and testing	Synthetic	Team-created; no third-party dataset licence claimed	N
Product-condition demonstration images	Google Gemini image-generation tools	Synthetic	Google Generative AI Additional Terms of Service	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT Plus and Pro	Architecture discussions, technical troubleshooting, code review guidance, presentation wording and declaration drafting	Major
Google Gemini 3.8 Flash	Generating illustrative images of damaged and undamaged products	Minor
Claude Opus 5.5, Claude Sonnet 5.5	Engineer and implement plans into features and fix those issues. Implement feedbacks from teammate and performing test	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated code, content and suggestions were reviewed by team members before being incorporated into the project. The team used automated backend tests and frontend end-to-end tests to check application behaviour. AI-generated extraction results are validated by the backend, and staff must confirm extracted information before it is used. Business decisions such as delivery pass/fail checks are made by deterministic application rules rather than AI-generated decisions. The team remains responsible for the final implementation and submitted work.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core access	Development and deployment of the SOPBox prototype.
RunPod	Compute credits or GPU access	Hosting and testing the Qwen3-VL model.
Krackeddev	AI API	Function calling through interaction with chatbox.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the deployment environment and PostgreSQL database, while RunPod provides GPU-based inference for image extraction. Both services could be replaced by alternative hosting, database and model-serving providers, although migration would require configuration and integration changes. The receiving rules, quantity checks, temperature checks and approval workflow are implemented in the application code and do not rely on AI to make business decisions.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

SOPBox collects account information such as usernames, display names, optional email addresses, roles and sign-in records. Business records include products, suppliers, deliveries, delivery-order and stock photos, stock counts, sales, comments, approvals and activity logs. Data is stored in Replit PostgreSQL, with workspace-scoped access controls and PostgreSQL Row Level Security. HTTPS, HttpOnly session cookies, password hashing and encrypted authenticator secrets are used for protection. Delivery-order photos are sent to the team's RunPod endpoint for inference and are not used for model training. Sales reports selected for AI reading and assistant questions may be sent to the separate KrackedDev AI service. Telegram and Google receive information when their optional integrations are used. Data is retained while a workspace remains open; after closure it is read-only for 30 days before deletion.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The vision-language model may misread handwritten, unclear, smudged or low-quality delivery documents, including product names, quantities and units. AI-generated outputs may also be incomplete or incorrect. A cold start on the serverless GPU can delay the first image extraction by a few minutes, and inference depends on RunPod availability and configuration. The second AI service also depends on external provider availability. SOPBox mitigates these risks by validating model responses, rejecting invalid extraction results, allowing staff to enter delivery lines manually, and requiring human confirmation. Temperature, pack conversion, purchase-order and approval decisions are handled by deterministic application rules. The assistant uses fixed tools and cannot independently approve or modify business records.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date] 1/11/2026